

Computer Networks I

Error Correction

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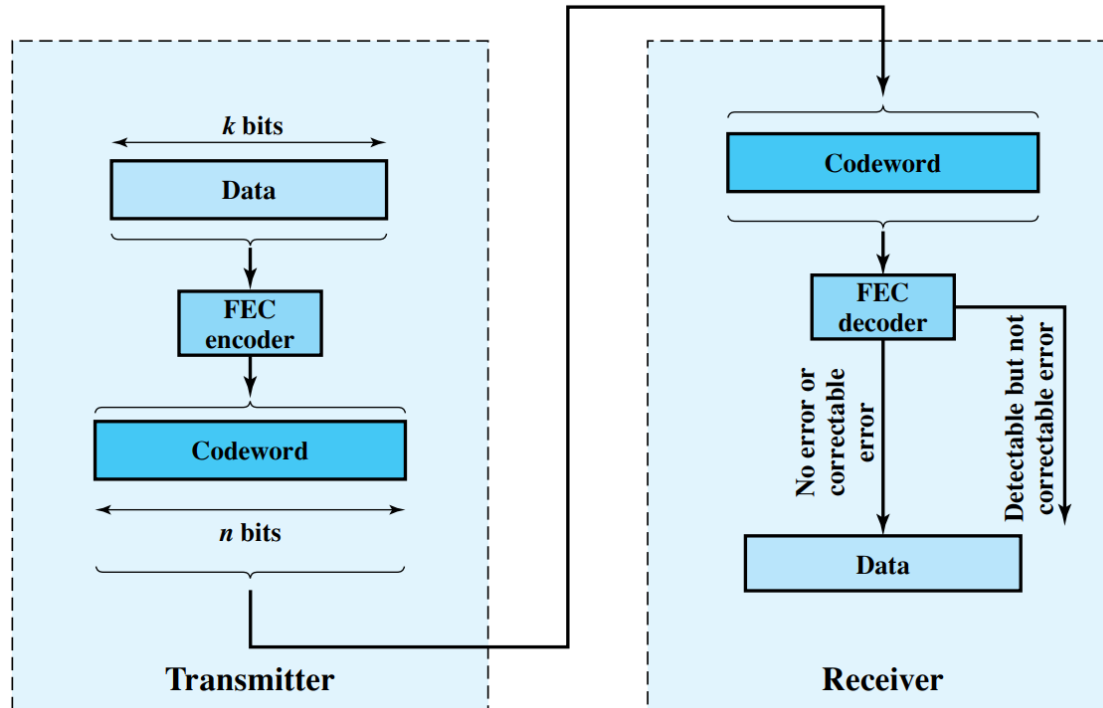
IIT Kanpur

Error Correction

- **Backward error correction:** Correction of detected errors usually requires data blocks to be retransmitted
 - Not appropriate for some wireless applications:
 - The bit error rate (BER) on a wireless link can be quite high, which would result in a large number of retransmissions
 - Propagation delay is very long compared to the transmission time of a single frame
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Error Correction

- **Forward error correction:** Need to correct errors on basis of bits received



Error Correction

- Hamming distance:
 - $d(v_1, v_2)$ – Hamming distance in between any two binary sequences v_1 and v_2
 - $d(v_1, v_2)$ is the number of bits in which v_1 and v_2 disagree
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Error Correction

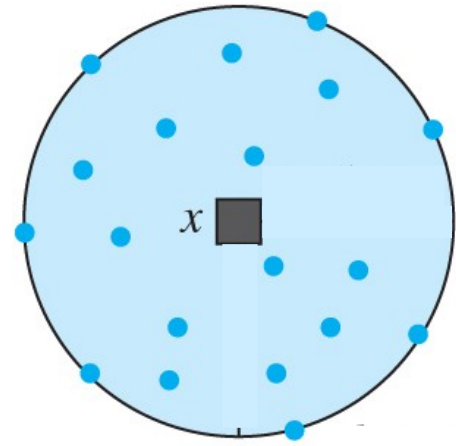
- Codeword of n bits = m message bits + r check bits
- Valid/legal and invalid/illegal codeword:
 - Total possible codewords with n bits is 2^n
 - All 2^m data messages are legal $\rightarrow$ each one of them has 1 legal codeword
 - All remaining codewords are invalid

0000000000

0000011111

1111100000

1111111111



Error Correction

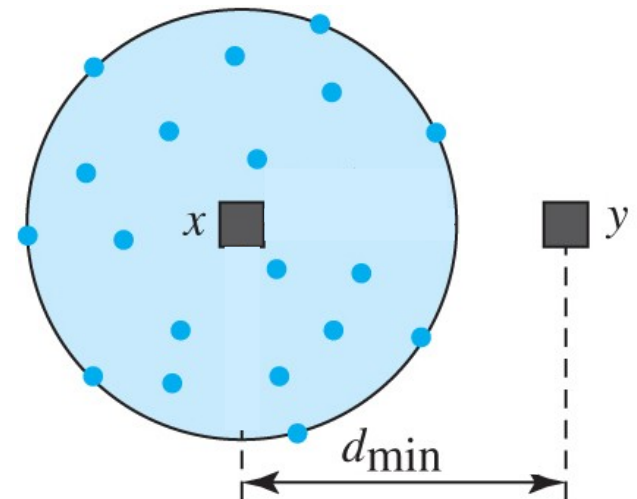
- Given an algorithm for calculating the check bits
 - We can construct the list of valid codewords
- **Hamming distance of the code:** Smallest Hamming distance in between any two valid codewords

0000000000

0000011111

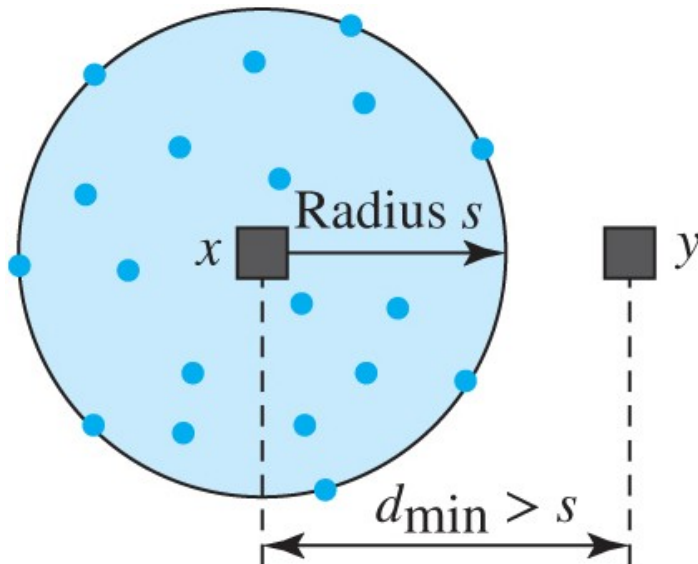
1111100000

1111111111



Error Correction

- To detect s bit errors, we need a distance $s + 1$ code
 - With such a code, there is no way that a s single bit error can change a valid codeword to another valid codeword

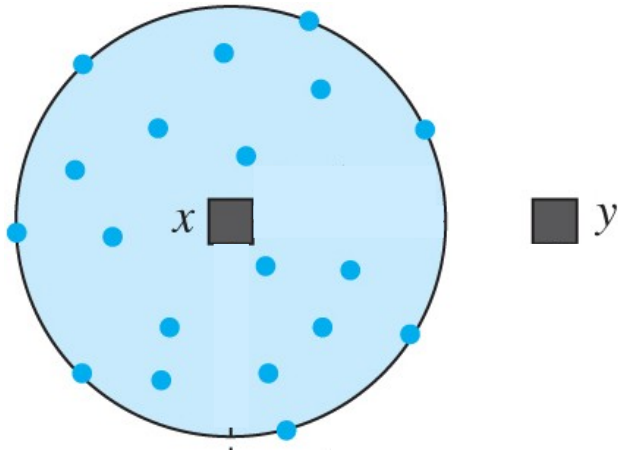


Legend

- Any valid codeword
- Any corrupted codeword with 1 to s errors

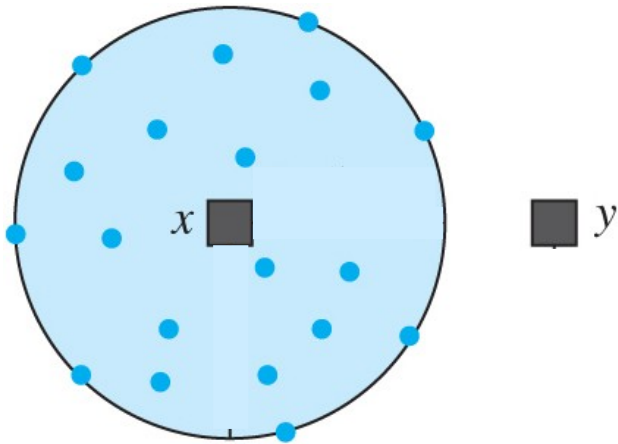
Error Correction

- To correct s bit errors, we need a distance $2s + 1$ code
 - With such a code, the legal codewords are so far apart that even s changes, the original codeword is still closer, than any other codeword



Error Correction

- To correct s bit errors, we need a distance $2s + 1$ code
 - Original codeword can be uniquely detected based on the assumption that a larger number of errors are less likely $\rightarrow$ time consuming search



Error Correction

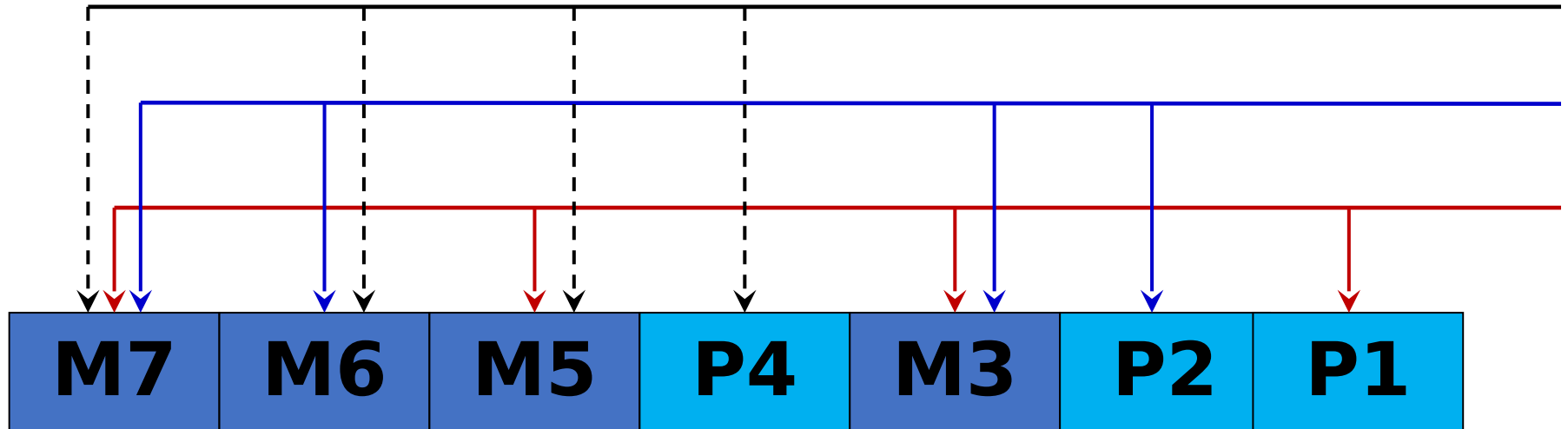
- Codeword of n bits = m message bits + r check bits
- We want to design a code that allows all single bit error to be corrected
 - Each of the 2^m legal messages $\rightarrow$ there is n illegal codewords at a distance 1 from it
 - As there are 2^n total number of bit patterns

$$2^m(n + 1) \leq 2^n \quad \therefore n + 1 \leq 2^{n-m} = 2^r \quad \therefore m + r + 1 \leq 2^r$$

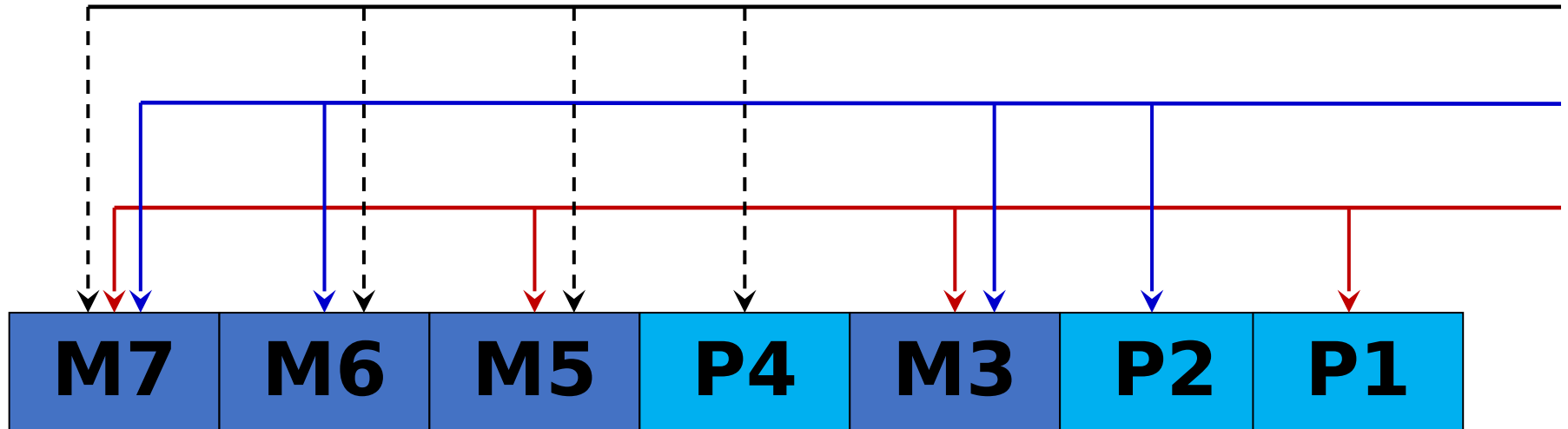
Error Correction

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 - So, given m , this put a lower bound on the check bits that are needed to correct single bit errors
 - For $m = 4$, $r = 3$
 - For $m = 7$, $r = 4$
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Hamming Code



Hamming Code



Error Correction Codes

- **Convolution codes:** GSM mobile phone system, satellite communications, 802.11
 - **Reed-Solomon code:** DSL, data over cable, satellite communications, CDs
 - **Low-density parity check:** Digital video broadcasting, Ethernet, 802.11
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Summary

□ Different error correction techniques discussed:

- Backward error correction
 - Forward error correction
 - Hamming distance
 - Hamming code
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